ANDRY HUANG



Senior Fullstack & AI Engineer (.NET | Next.js | Node.js | RAG / AI Agents)

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PROFESSIONAL SUMMARY

Senior Fullstack & AI Engineer with 7+ years of experience. Spent 3+ years remote at MRI Software (Singapore) building enterprise leasing and tenant-management SaaS, including production RAG features on OpenAI and pgvector. Founded Qualiv, an AI recruitment SaaS, and built ERP, POS, and CMS systems for businesses in Surabaya. Core stack: .NET Core, Next.js, Node.js, NestJS, SQL tuning, and LLM/RAG.

TECHNICAL SKILLS

• **Backend & Cloud Architecture**: .NET Core, C#, Node.js, NestJS, Laravel, Java Spring Boot, Python (FastAPI), REST & GraphQL APIs, SaaS Architecture
• **AI Engineering & Automation**: RAG Architecture, OpenAI GPT-4o, Google Gemini API, Voice AI (STT/TTS), Anthropic Claude API, Model Context Protocol (MCP), Autonomous AI Agents, pgvector, BullMQ, Prompt Engineering
• **Frontend & Mobile**: Next.js, React, TypeScript, JavaScript, Tailwind CSS, Flutter, Android Native, Livewire, Alpine.js
• **Databases & Storage**: PostgreSQL, SQL Server, MySQL, Oracle DB, Redis, Elasticsearch, AWS S3, Azure Blob, Cloudflare R2
• **DevOps, CI/CD & Security**: Docker, GitHub Actions, Azure DevOps, GitLab CI, Linux Administration, Nginx, Fail2ban, SonarQube, SSH Security
• **Testing & Integrations**: Playwright, Selenium, Midtrans Payment Gateway, RajaOngkir, RFID Hardware, Leaflet GIS

WORK EXPERIENCE

Fullstack Developer | MRI Software (formerly Anacle Systems Ltd) (Singapore, Remote) | Nov 2022 - Mar 2026

- Built features for Anacle Simplicity and the SMRT Tenant Management System through go-live: mall tender submissions, space leasing, tenant applications, and daily operations. Stack: .NET Core, .NET Framework 4.8, Next.js, NestJS.
- Shipped production RAG pipelines on the OpenAI API and PostgreSQL (pgvector) for semantic document search and automated executive reports.
- Wrote a microservice that migrated 100,000+ files to SharePoint (Graph API), AWS S3, and Azure Blob Storage, with RBAC and audit logging.
- Refactored legacy stored procedures and wrote regression tests in Selenium and Playwright for core SaaS modules.
- Led the SonarQube rollout across enterprise repositories, fixed the security vulnerabilities it flagged, and set CI/CD quality gates that block deployments with new vulnerabilities.

Freelance Fullstack & AI Engineer | Self-employed (Surabaya, Remote) | 2021 - Present

- Build AI products end to end, from architecture to production: Qualiv, an AI recruitment SaaS, and an AI tire-recommendation assistant for PT Nagamasban.
- Build POS and inventory systems for retail stores (yarn, RFID jewelry, grocery, tire), plus the Knit and Cro online shop and the Wonderful Works website and CMS.

Fullstack Developer | Software House (Surabaya, Indonesia) | Mar 2022 - Nov 2022

- Maintained a legacy enterprise Tool Management System (TMS) in Java, Spring Boot, and Oracle. Led a Java version upgrade, a large refactor, and a code-review process, and optimized complex SQL queries.
- Tracked down the edge-case business-logic bugs that were corrupting data, patched them, and repaired 10,000+ corrupted records.
- Maintained self-hosted Git servers and set up Elasticsearch clusters for log indexing.
- Traced recurring high-priority incidents to backend bottlenecks and fixed them, cutting incident tickets quarter over quarter.

Full-Stack Developer | PT Pabrik Kertas Tjiwi Kimia Tbk (Surabaya / Mojokerto, Indonesia) | Sep 2019 - Feb 2022

- Built manufacturing dashboards and reports for pulp & paper production in .NET Core, C#, ASP Classic, and SQL Server. Query tuning cut long-running reports from minutes to seconds.
- Automated plant reporting with a WhatsApp bot (Python, Java, Selenium) that sends scheduled dashboard snapshots of OEE and HWT production metrics.
- Shipped a native Android barcode scanner app for warehouse stock-in and stock-out.
- Developed internal apps, including a company-wide daily COVID-19 health check-in and a digital company raffle draw.

FEATURED PROJECTS

• **Qualiv | AI Recruitment SaaS Platform** (Founder & Lead Architect): Multi-tenant recruitment SaaS, built from scratch. It parses CVs (PDF/DOCX), runs AI chat interviews, logic tests, and spoken video interviews in Indonesian, and scores each candidate against the job's requirements. In the video interview, Gemini transcribes each spoken answer and writes the next question, and OpenAI text-to-speech reads it aloud over a WebSocket. Companies pay for subscriptions through Midtrans. Tech Stack: Next.js, NestJS, OpenAI GPT-4o, Google Gemini, Speech-to-Text & TTS, Socket.IO, PostgreSQL, Prisma, BullMQ, Redis Queue, Cloudflare R2, Midtrans

• **PT Nagamasban | AI Sales Assistant & Catalog** (Lead AI Architect): Recommendation and sales-training tool for a Surabaya tire distributor. Matches tires to a customer's vehicle and budget using OpenAI embeddings and semantic search. Tech Stack: Next.js, NestJS, OpenAI Embeddings, Semantic Search, Tailwind CSS

• **Retail POS & Inventory Systems** (Fullstack Architect): POS and inventory systems for four stores: Knit and Cro Pluit yarn store (color and dye-lot variants, stock opname, role-based access), Wijaya Mas jewelry store (RFID tagging, daily gold-price calculation, buyback), Tiga Putra grocery (PO/SO, FIFO expiry tracking), and Cahaya Terang tire shop (sales and purchasing with a double-entry A/R and A/P ledger). Tech Stack: Laravel 11, Livewire 3, MySQL 8, Alpine.js, RFID Hardware, RBAC

• **Wonderful Works | Architecture & Contractor Platform** (Fullstack Architect): Company website and CMS for a Surabaya architecture and contracting firm. Staff can edit pages in a drag-and-drop composer and manage bilingual (ID/EN) content, project photos, and users from an admin panel with JWT auth. JSON-LD for SEO; deploys through Docker and Jenkins. Tech Stack: Next.js 16, React 19, Tailwind CSS v4, Prisma, PostgreSQL, Puck Visual Editor, Docker, Jenkins

• **Knit and Cro | Online Shop** (Fullstack Architect): Online shop for a yarn store in Pluit, Jakarta, connected to the store's inventory. Customers pay through Midtrans (QRIS, bank transfer, e-wallet) and get shipping rates from RajaOngkir; staff handle orders, tracking numbers, and returns from an admin dashboard. Tech Stack: Laravel 11, Livewire 3, MySQL 8, Midtrans, RajaOngkir API, Tailwind CSS

EDUCATION

Bachelor of Science in Computer Science (S1 Teknik Informatika) | **University of Surabaya (UBAYA)**, Indonesia (2015 - 2019) | GPA: 3.60 / 4.00 (Cum Laude)

CERTIFICATIONS

• **Anthropic AI Certifications Suite (9 Certifications)** (Anthropic, 2026): **Building with Claude API** | **Claude Code in Action** | **Intro to MCP** | **MCP Advanced** | **Agent Skills** | **Subagents** | **AI Capabilities & Limitations** | **Claude Cowork** | **Claude 101**

• **AI Agents with Model Context Protocol**: **Vanderbilt University** (**Coursera Verification**, 2026). Hands-on course covering agent design, tool use, and building MCP servers and clients.

• **Software Architecture & Design of Modern Large Scale Systems**: **Udemy** (**Udemy Verification**, 2026). Covers microservices, caching, and fault-tolerant design for high-throughput systems.

LANGUAGES

• English (Professional Working Proficiency), Indonesian (Native Proficiency)